

Consider following lock algorithm:

```
class Filter implements Lock {
    int[] level;
    int[] victim;

    public Filter(int n) {
        level = new int[n];
        victim = new int[n]; // use 1..n-1
        for (int i = 0; i < n; i++) {
            level[i] = 0;
        }
    }

    public void lock() {
        int me = ThreadID.get();
        for (int i = 1; i < n; i++) { //attempt level 1
            level[me] = i;
            victim[i] = me;
            // spin while conflicts exist
            while ((∃k != me) (level[k] >= i && victim[i] == me)) {};
        }
    }

    public void unlock() {
        int me = ThreadID.get();
        level[me] = 0;
    }
}
```

7.2.1

Prove that Filter Lock algorithm satisfies mutual exclusion.

▷

Let's use Lemma 2.4.1 from 'The Art of Multiprocessor Programming', which states: For j between 0 and $n - 1$, there are at most $n - j$ threads at level j .

The last level, $j = n - 1$, is the critical section. Therefore, by Lemma 2.4.1, the critical section level will have at most $n - (n - 1) = 1$ threads.

Hard way - getting a contradiction, assuming that two threads entered level $n - 1$.

Let threads A and B both are on the level $n - 1$. Let A be the one that last wrote to $victim[n - 1]$.

Then, both must have set level to $n - 1$ and completed the loop on $i = n - 1$. Then, the order of operations is:

```
B.write(level[B] = n - 1) -> B.write(victim[n - 1] = B) -> A.write(victim[n - 1] = A)
-> A.read(level[B])
```

Therefore, when spinning, A sees $level[B] \geq n-1$ and $victim[n-1] == A$, and it should keep spinning. Contradiction.

◁

7.2.2

Prove that Filter Lock algorithm is starvation-free.

▷

Induction with levels, starting from level $n - 1$.

Base case: trivial.

Transition: Assume all threads that reach level $j + 1$ or higher eventually enter the critical section.

Let thread A is stuck at level j . By the induction hypothesis, eventually no thread will remain at levels $> j$. Once A sets $level[A] = j$ any thread from lower levels won't enter j .

Now induction on number of threads: Base case is trivial. if there are many threads stuck on a level, only one of them is spinning, and other ones will advance. The number of stuck threads therefore decreases for at least one, eventually reaching 1, base case.

◁